

Multi-view semi-supervised classification overview

Lekang Jiang

Department of Computer Science, University of Nottingham, Ningbo, 315000, China, scylj1@nottingham.ac.uk,
Corresponding author

ABSTRACT

With the continuous exploration of data collection technology, the multi-view data, obtained from different ways or levels for the same object, and the multi-view algorithm, used for modeling and solving problems from multiple perspectives, have attracted extensive attention. In addition, if the traditional supervised learning method is used in multi-view learning, a great amount of labeled data are required to ensure the accuracy of the training model, but it is difficult to obtain the labeled samples. Thus, combining semi-supervised learning with multi-view method can effectively solve the difficulties of labelling. In this paper, we will describe the related multi-view semi-supervised learning algorithms and development status in the aspect of classification, and then introduce the latest real world applications. Finally, we will summarize and point out the possible research directions in the future.

CCS CONCEPTS

• **Computing methodologies** → Modeling and simulation; Model development and analysis.

KEYWORDS

Multi-view learning, semi-supervised learning, classification

ACM Reference Format:

Lekang Jiang. 2021. Multi-view semi-supervised classification overview. In *2021 2nd International Conference on Artificial Intelligence and Information Systems (ICAIS '21)*, May 28–30, 2021, Chongqing, China. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3469213.3470387>

1 INTRODUCTION

With the development of data collection technology, the forms of data become more diverse. Samples may not be represented comprehensively from a single view. For instance, both the text information in a web page and the anchor text linking to this page could be used to represent the page in web page classification problems. Another example is advertisement images classification. An advertisement picture could be identified by the content and colors, or its title and text information. Different features of the same sample data from different perspectives can be called multi-view data. These

data are not only different but also interrelated, which show the features more comprehensively from multiple angles. Therefore, it is significant for improving the performance to make full use of the advantages of multi-view data in machine learning.

There are two basic learning algorithms in multi-view learning process: supervised learning and unsupervised learning. In supervised learning, a large number of labeled data is used to train and establish appropriate models for prediction and analysis of other samples. However, supervised learning needs much labeled data to ensure the accuracy of the model, but it requires both manpower and material resources to obtain those labeled samples. In fact, with the development of data collection technology, it is easier to obtain unlabeled samples. Therefore, semi-supervised learning, using a few labeled data and many unlabeled data to train models, has attracted extensive focus in machine learning research in recent years.

In previous studies, most reviews focused on either semi-supervised learning or multi-view learning. For example, Tang et al. [1] summarized the three main methods of multi-view learning, named Co-training, multiple kernel learning and subspace learning. Han et al. [2] divided semi-supervised learning into six aspects according to different learning tasks: classification, clustering, regression, dimension reduction, imbalanced data classification and noise reduction. However, paper about the combining of multi-view learning and semi-supervised learning is little. Therefore, this paper will review the multi-view semi-supervised learning, which combines both multi-view learning and semi-supervised learning. It will first introduce the related algorithms and development status in classification, then illustrate the latest real world applications, and finally make conclusion and prospect. This paper aims to introduce the related algorithms of multi-view semi-supervised learning, so that readers can understand the research status and development trend of this new algorithm, and pave the way for future studies.

2 MULTI-VIEW SEMI-SUPERVISED CLASSIFICATION ALGORITHMS

Multi-view learning can make full use of the different features in data sets to improve learning performance. In the process of view selection, each view usually contains unique data features, so that the information in each view can complement each other. Thus, it describes the data more comprehensively and accurately to get a more optimized model. At the same time, the consistency between different views should be maximized, so that the performance of the model could be improved by comparing the classification results in multiple perspectives. In the process of multi-view classification, if the traditional supervised learning method is adopted, the problem of data labelling often exists. Therefore, the combination of multi-view and basic semi-supervised learning can not only explore deep features of data, but also improve the utilization rate of unlabeled data to reduce the cost of multi-view learning.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

ICAIS '21, May 28–30, 2021, Chongqing, China

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-9020-0/21/05...\$15.00

<https://doi.org/10.1145/3469213.3470387>

Table 1: COMPARATION AMONG SEMI-SUPERVISED LEARNING ALGORITHM

Name	Application scope	Advantages	Disadvantages
Generative learning algorithm	Features are simple, low dimensional	Simple and intuitive	Difficult to solve for large-scale data problem
Semi-supervised support vector machine	Suitable for most cases, especially for small sample data set	Performance better than traditional SVM	Not so efficient and performance bad in some specific cases
Graph based semi-supervised algorithm	Suitable for most cases, especially for image and language processing	Intuitive, good interpretation and learning performance	Graph is difficult to build for large-scale data
Multi-view semi-supervised learning	Suitable for most cases, especially multi-view data	Can combine with multiple classifiers to improve performance	Easy to introduce noise

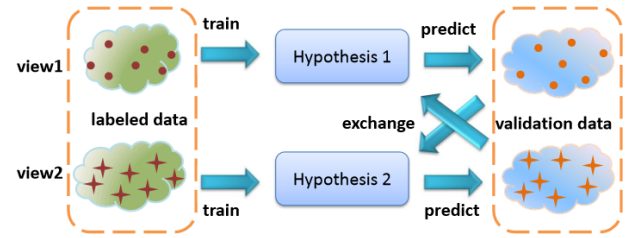
2.1 Basic semi-supervised classification

The basic semi-supervised classification algorithms include generative learning algorithm (GLA), semi-supervised support vector machine (S3VM) and graph based semi-supervised algorithm. The multi-view semi-supervised learning algorithm can combine with other basic semi-supervised classification algorithms to improve the generalization ability of the model by using the differences of multiple learners. These basic learning frameworks are briefly introduced below and summarized in Table 1. In generative learning algorithm (GLA) [3-5], samples and class labels are considered to be some specific probability distribution. And labels with the highest posterior probability distribution are found and tagged based on generative model. Semi-supervised support vector machine (S3VM) [6-10] trains SVM with labeled data, estimates the label of unlabeled data, and updates the predictive model with iteration, for example Transductive Support Vector Machine (TSVM) [6] and Laplacian Support Vector Machine (Laplacian SVM) [7]. The graph based algorithm first establishes a graph based on labeled and unlabeled data. The nodes in the graph correspond to the samples, and the edges correspond to the similarity between the examples. Then, according to the adjacency relationship, unlabeled data are affected by labeled data to obtain the optimal model parameters [11-16], for example, discrimination-aware manifold regularization [15] and higher order Laplacian regularization method [16].

Semi-supervised learning algorithm has been greatly improved in learning performance and efficiency, but each method has its own limitations. The biggest advantage of multi-view semi-supervised learning is that it can combine the advantages of various classifiers, complement each other and further improve the accuracy and efficiency of the model. Therefore, multi-view semi-supervised learning has become the focus recently in machine learning area.

2.2 Multi-view semi-supervised classification

1) *Based on Co-training algorithm* Blum and Mitchell [17] first proposed the Co-training algorithm for multi-view semi-supervised learning problem, as shown in Figure 1 [18]. The algorithm trains two different classifiers from two different perspectives by using the given data set. In the process of each co-training iteration, the two classifiers are used to train the unlabeled data, and find the data with high confidence. These data are added to the training set for the next iteration, which maximizes the prediction consistency

**Figure 1:** Process of Co-training algorithm

of the classifiers from the two perspectives on the verification set and improves the learning performance. However, the Co-training algorithm needs to satisfy the premise that there are two sufficient and redundant perspectives. Sufficiency means that each perspective is enough to learn a good classifier. Redundancy means that the two perspectives are independent of each other under the condition of given label. However, in most practical applications, the above assumptions are not always satisfied. As a result, a series of improvement and expansion on Co-training algorithm and theory have emerged. The main development trend of Co-training algorithms is shown in Table 2

Collins and Singer [19] improve the Co-training algorithm and optimize the objective function to accurately measure the consistency among different perspectives. Nigam and Ghani [20] analyze the effectiveness and applicability of Co-training. It is proved that when the attribute set is sufficiently large, it can achieve good results by dividing the attribute set into two perspectives randomly for Co-training. Goldman and Zhou [21] propose a Co-training algorithm based on noise learning theory [22], in which sufficient redundant views are not required and multiple different kinds of classifiers are available. Muslea et al. [23] propose CO-EMT algorithm based on active learning and CO-EM algorithm, which perform better than Co-training or CO-EM algorithm, and have better robustness. Brefeld and Scheffe [24] use SVM instead of Naive Bayesian classifier to make the algorithm support more classifier. Sindhwani et al. [25] propose a collaborative regularization algorithm, which is non greedy and includes convex cost function. Balcan et al. [26] do further study on Co-training, and find that if the data distribution satisfies the expansibility hypothesis, then strong learners

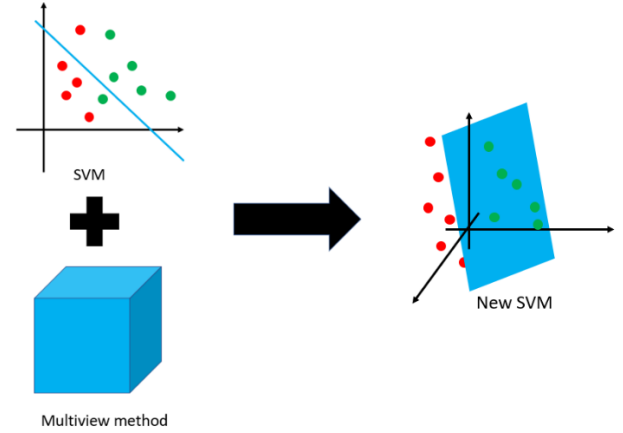
Table 2: MAIN DEVELOPMENT TREND OF CO-TRAINING ALGORITHM

Main development trend
Co-training is first proposed
Improve the performance of Co-training algorithm
More classifiers are supported in Co-training
Neither sufficient nor redundant views are required
Co-training can be used for single view data
Many other theories are combined with Co-training

can be trained on each perspective. Zhou and Goldman [27] expand the Co-training algorithm to make the algorithm support other classifier. Zhou and Li [28] propose Tri-training algorithm, in which neither sufficient redundant views nor different types of classifiers are required. Then, they propose Co-Forest algorithm based on Tri-training, which has better performance in integration learning [29]. Wang and Zhou [30] study Co-training based on PAC classifier, and find that different classifiers also can be used to do Co-training successfully for single view data. Sun [31] uses canonical correlation analysis (CCA) [32] in Co-training to check unlabeled samples, which improve the reliability of model. Yu et al. [33] propose Bayesian Co-training, which avoids local maximum, and estimates the reliability of each view automatically to adapt to noisy or unreliable views. Sheng et al. [34] propose a Co-training algorithm based on rough sets theory [35], which improving the effectiveness and efficiency of Co-training model in risk decision making environment. Guo et al. [36] propose confidence estimation for semi-supervised learning based on graph (CESL). They improve the efficiency of Co-training algorithm for unlabeled data selection, and reduce the introduction of noise. Guo et al. [37] take advantage of the optimization function of genetic algorithm to generate better classification rules, to reduce the noise, and improve the accuracy and performance of classifier. Chen et al. [38] introduce the multi-view theory, coding theory and gravity formula into Co-training classification algorithm. It can prevent the error accumulation in the iterative process and improve the generalization performance of the classification model. Guo et al. [39] proposed a method that the samples with inconsistent labels are deleted dynamically, by comparing the label predicted by the classifier and its initial label. Better generalization ability and faster convergence speed are obtained in this method. Jing et al. [40] proposes a Semi-supervised Multi-view Feature Learning (SMFL), used for web page classification, which improve the accuracy and efficiency. Chen et al. [41] proposed a Multi-view semi-supervised learning for classification on dynamic networks (MSCD) to have better results in classification.

2) Based on Support Vector Machine

In addition to Co-training algorithms, there are also some multi-view semi-supervised classification methods based on support vector machine. Semi-supervised vector machine (S3VM) uses labeled data to train SVM, estimate the label of unlabeled data, and updates the predictive model iteratively. It performances better than traditional SVM and can be applied to high-dimensional data, but it also has some problems. For example, when the loss function is not convex, local minimum point may be found instead of minimum point, which might lead to a lower generalization ability. If the

**Figure 2: Schematic diagram of SVM based algorithm**

multi-view method is added to the support vector machine and improved to a new classification vector machine, as shown in Figure 2, the shortcomings of the traditional SVM can be effectively improved. Thus, the utilization of data and model accuracy could be further improved.

Farquhar et al. [42] propose the SVM-2K model combining KCCA [43] and SVM to improve the algorithm's performance. Sun [44] propose a sparse semi-supervised learning framework, which is applicable to the methods such as multi-view co-regularized least squares and single-view Laplacian support vector machines (SVMs), which accelerates the process of obtaining the predicted results. Sun [45] propose multi-view Laplacian support vector machines (SVMs), which combines manifold regularization and multi-view regularization with SVMs. This method uses optimization function in a finite-dimensional Euclidean space instead of in a reproducing kernel Hilbert [46] space. Li et al. [47] propose a novel two-view transductive SVM, which improves the classification performance by taking advantage of many unlabeled data and their different representations. Xie et al. [48] propose a multi-view Laplacian twin support vector machine, and it is further extended to kernel multi-view Laplacian twin support vector machine. Brefeld [49] propose a semi-supervised hidden Markov perceptron and a semi-supervised hidden Markov support vector learning algorithm, based on the principle of maximizing the consensus among multiple independent hypotheses to improve model accuracy.

3) Others

Apart from Co-training based and SVM based method, there are also some other multi-view semi-supervised classification method. Zhou et al. [50] propose a method that work on double view environment, by using canonical correlation analysis from two perspectives for semi-supervised learning. They find that in the case of few data, only one labeled training instance is enough for semi-supervised learning. Rosenberg [51] proposed the multi-view point cloud regularization (MVPCR), which summarizes some semi-supervised kernel methods. Luo et al. [52] develop multi-view vector-valued manifold regularization, used to integrate multiple features of data sets.

3 LATEST APPLICATIONS

In recent years, with the development of multi-view semi-supervised learning, it is widely used in various fields. This section will introduce the latest research progress from four aspects: natural language processing, image recognition, behavior identification and other applications.

3.1 Natural language processing

In natural language processing, a large number of high-quality labeled corpuses are often needed for learning, but it is very difficult to obtain labeled samples. Semi-supervised method can be used to learn a large number of unlabeled corpuses, which reduces the difficulty of training and improves the performance of the model. The multi-view method can fully explore the features of the corpus, learn from each other from multiple perspectives, and further improve the accuracy of the model. This paper introduces the application of multi-view semi-supervised learning in natural language processing from three aspects: machine translation, spam comment recognition and emotion recognition.

1) Machine translation

Multi-view semi-supervised learning is helpful to improve the performance of machine translation. Hu et al. [53] proposed a relationship classification method based on bilingual Co-training, which makes full use of the complementarity between languages to reduce the demand for large-scale training data, improves the performance of relationship classification between Chinese and English, and reduces the need for labeling training data. Li et al. [54] proposed a bilingual maximum noun phrase recognition algorithm based on semi-supervised learning, by taking advantage of the mutual translation and recognition complementarity of Chinese and English maximal noun phrase. It improved the poor consistency of maximum noun phrase recognition and weak ability cross different domain recognition. Huang et al. [55] optimized the traditional machine translation by using Co-training method, and applied it to neural machine translation tasks. It improved the translation quality of the original neural machine translation model, especially when the number of corpus was scarce. In addition, multi-view semi-supervised learning can also improve the performance of phrase mining in specific fields. For example, in the field of e-commerce, text usually does not follow the regular expression rules. Therefore, Xu et al. [56] proposed a Co-training based phrase mining method for e-commerce field. It can effectively detect specific expressions in e-commerce domain text through the phrase classification model based on semantic features. It reduces the cost of labeling training

data in corpus, and improve the overall mining performance of the model with higher precision and recall.

2) Spam comment recognition

Li et al. [57] constructed the AdaBoost classifier and support vector machine classifier, and used Co-training algorithm to identify spam comment, which improved the classification accuracy and reduced the work of sample labeling. Then, aiming at solving the problem that it is difficult to obtain large-scale labelled data sets, Du et al. [58] proposed a spam comment recognition method combining PCA with Co-training algorithm, which improved the recognition performance. At the same time, PCA can reduce the redundancy of original samples and improve the classification efficiency of the model. This method uses PCA to extract features and eliminate the correlation between two independent views: comment text and comment behavior. Finally, according to the differences between different views, appropriate classifiers are selected for Co-training to complete the recognition of spam comments.

3) Emotion recognition

The text sentiment classification method based on machine learning has achieved good results, but it needs a great quantity of manpower and material resources to obtain labeled data in practical application. Li et al. [59] introduced multi-view semi-supervised method into text sentiment classification method to solve the problem of using unlabeled samples. Li Yao [60] used multi-view semi-supervised learning to study the microblog emotion classification, which improves the accuracy of classification and reduces the demand for sample labelling. When judging the emotional polarity of a document, different words and sentences have different contribution. Therefore, metaphors and other key sentences which contain more emotional words and summary are extracted from the whole document. The emotional word attribute, position attribute, punctuation attribute and keyword attribute of the sentence are fully considered to improve the classification performance [61].

3.2 Image recognition

1) *Image classification* Improving the accuracy of image classification is the key research direction of multi-view semi-supervised classification. Li [62] introduced Hessian regularization into collaborative training. When the number of initial labeled samples is small, Hessian regularization can use a large number of unlabeled sample information to explore the intrinsic geometric characteristics of data edge distribution, and achieves good recognition effect in scene recognition. At the same time, the collaborative density peak spectrum clustering algorithm model is proposed, and the multi-view semi-supervised clustering algorithm is established, which is applied to action recognition and image recognition to improve the clustering accuracy. Yuan et al. [63] proposed a semi-supervised classification method for polarimetric SAR images based on Co-training and ensemble learning. The method uses support vector machine as the base classifier of semi-supervised learning, and effectively combines the features under the decomposition of multiple polarimetric targets through collaborative learning mechanism. It improves the generalization ability and the accuracy, also when labeled samples are few. Peng et al. [64] proposed a multi-view semi-supervised method based on depth segmentation model. The

model is trained on a subset of labeled data, and the unlabeled images are used to exchange information to improve the performance of semantic image segmentation in semi-supervised environment, while maintaining the diversity of information.

2) 3D image segmentation

Image segmentation is a process of dividing an image into several parts with unique properties, which is a significant procedure before image processing and image analysis. Reinbacher et al. [65] propose a 3D segmentation method based on the color, edge and spatial information of numbers of views of an image, which achieves the accurate segmentation of pixels. This method is faster and more accurate than the traditional 2D methods. Xia et al. [66] propose Uncertain perceptual Multi-view Co-training (UMCT) for medical image segmentation. This 3D multi-view method not only improves the utilization of unlabeled images, but also improves the segmentation accuracy, which shows a strong potential in practical applications. Sun et al. [67] propose a semi-supervised 3D pulmonary nodule segmentation network for segmentation of nodes in CT images. It solves the problem of low segmentation accuracy in traditional view, and improves the performance of the algorithm when the data set is insufficient.

3.3 Behavior identification

Human behavior identification is widely used in video surveillance, intelligent transportation and many other fields. However, in the real world scene, complex background, object changes and other problems will lead to the reduction of recognition accuracy and difficulty in obtaining training samples. Wang et al. [68] applied Tri-training method to study pedestrian based on the existing classification methods and combined with Co-training framework, which reduced the impact of insufficient training samples to a certain extent, improved the accuracy and reduced the learning time. Tang et al. [69] proposed a human behavior recognition method based on multi learner Co-training model, which can reduce the required labeled samples and improve the generalization ability. Wang et al. [70] combined the abstract characteristics, autonomous behavior characteristics and interactive information features of learners, and then used multi-view semi-supervised method to label the learners' behavior data. Finally, judging whether the learners were pseudo learners according to the labeling results. Liu et al. [71] introduced Laplacian regularization into the Co-training model, using a large number of unlabeled sample data to train two classifiers from different perspectives. The unknown information was exchanged between the two classifiers and the classifiers were updated to improve the accuracy of behavior recognition.

3.4 Other applications

In addition to the above applications, multi-view semi-supervised learning has also been applied to many other fields. For example, when the number of labeled samples is small, multi-view semi-supervised learning is used to improve the performance of phone APP classification [72]. In the field of engineering, a semi-supervised machine learning method with two classifiers is used to improve the accuracy of fault classification in power transmission and distribution system [73]. In the medical field, the multi-view

semi-supervised training is applied to miRNA prediction, and maximize the multi-view learning advantage [74]. Without any new labeled training data, it can significantly improve the classification accuracy of the sequence based and expression based classifiers, which shows a great application prospect, especially for the species which have less training data.

4 CONCLUSION AND PROSPECT

With the diversification of data forms, semi-supervised learning using multi-view method can fully mine the features of data and reduce the labeled data needed for training. In addition, a large number of unlabeled data are used for learning, which effectively reduces the demand of human and material resources, and the multi perspective features also improve the generalization ability of the model. This paper summarizes the multi-view semi-supervised classification algorithm, then, its application in various fields are introduced. On the basis of the existing research status, this paper makes the following prospects for the future research direction:

For the quality of multi-view data, not all features are valuable and worth studying, and there will be noise and redundancy in most cases. Therefore, optimizing multi-view data in the process of data collation, or eliminating noise and redundancy in the training process, have a significant impact on improving the performance of the algorithm.

For the accuracy of the algorithm, when the false labels provided by different classifiers are wrong, error labels will appear in the new training set again, which will reduce the performance of the model. Secondly, there are many parameters involved in multi-view semi-supervised learning, but the experience of parameter selection is not rich. Therefore, how to reduce the false labeling error, how to select the required parameters or to reduce the parameters are worthy of further study.

For the efficiency of the algorithm, there are many researches on making full use of information and improving the accuracy of the model, but the efficiency of the algorithm is always a focus worthy of attention. How to improve the accuracy, reduce the running time, and further optimize the algorithm, is worth in-depth study.

For the algorithm verification, some multi-view semi-supervised algorithms are only verified in specific data sets, lacking of theoretical basis and proof in many practical applications. Therefore, how to use a large number of different data sets to verify and improve the stability and effectiveness of the algorithm in practical application has significance to study.

For the practical application of the algorithm, the research and application of multi-view semi-supervised classification has made great progress, but the research on multi-view semi-supervised clustering and regression is relatively small. There is still a wide research space, closely due to many difficulties in its research. Therefore, the theory and practical application of multi-view semi-supervised clustering and regression will be important for the future research.

REFERENCES

- [1] Tang JJ, Tian YJ. A review of multi-perspective learning [J]. Mathematical Modeling and its Applications, 2017, 6(003):P.1-15.
- [2] Han Song, Han QiuHong. A review of semi-supervised learning research [J]. Computer Engineering and Applications, 2020, 056(006):19-27.

- [3] B. Shahshahani, D. Landgrebe. The effect of unlabeled samples in reducing the small sample size problem and mitigating the Hughes phenomenon. *IEEE Transactions on Geoscience and Remote Sensing*, 1994, 32(5): 1087-1095.
- [4] D. J. Miller, H. S. Uyar. A mixture of experts classifier with learning based on both labelled and unlabelled data. In: M. Mozer, M. I. Jordan, T. Petsche, eds. *Advances in Neural Information Processing Systems 9*, Cambridge, MA: MIT Press, 1997, 571-577.
- [5] K. Nigam, A. K. McCallum, S. Thrun, T. Mitchell. Text classification from labeled and unlabeled documents using EM. *Machine Learning*, 2000, 39(2-3): 103-134.
- [6] Chen Y, Wang G, Dong S. Learning with progressive transductive support vector machine [J]. *Pattern recognition letters*, 2003, 24(12):p.1845-1855.
- [7] Lee S, Park C, Koo J Y. Feature selection in the Laplacian support vector machine[J]. *Computational Stats & Data Analysis*, 2011, 55(1):567-577.
- [8] Cevikalp H, Franc V. Large-scale robust transductive support vector machines [J]. *Neurocomputing*, 2017, 235:199-209.
- [9] Yang Z, Xu Y. A safe screening rule for Laplacian support vector machine[J]. *Engineering Applications of Artificial Intelligence*, 2018, 67:309-316.
- [10] Liu Z, Liu H, Zhao Z. Weighted least squares support vector machine for semi-supervised classification[J]. *Wireless Personal Communications*, 2018, 103(1):797-808.
- [11] A. Blum, S. Chawla. Learning from labeled and unlabeled data using graph min-cuts. In: *Proceedings of the 18th International Conference on Machine Learning (ICML'01)*, San Francisco, CA, 2001, 19-26.
- [12] Kim K, Tompkin J, Pfister H, et al. Context-guided diffusion for label propagation on graphs[C]// *Proceedings of the IEEE International Conference on Computer Vision*, 2015:2776-2784.
- [13] Wang X L, Lin H S. Label propagation through minimum cost path[J]. *Chinese Journal of Computers*, 2016, 39(7):1408-1418.
- [14] Xiu Y, Shen W, Wang Z, et al. Multiple graph regularized graph transduction via greedy gradient max-cut[J]. *Information Sciences*, 2018, 423:187-199.
- [15] Wang Y, Chen S, Xue H, et al. Semi-supervised classification learning by discrimination-aware manifold regularization[J]. *Neurocomputing*, 2015, 147:299-306.
- [16] De Sousa A R, Batista G E. Robust multi-class graph transduction with higher order regularization[C]// *2015 International Joint Conference on Neural Networks (IJCNN)*, 2015: 1-8.
- [17] Blum, T. Mitchell. Combining labeled and unlabeled data with co-training. In: *Proceedings of the 11th Annual Conference on Computational Learning Theory (COLT'98)*, Wisconsin, MI, 1998, 92-100.
- [18] Xu C, Tao D, Xu C. A survey on multi-view learning [J]. *arXiv preprint arXiv:1304.5634*, 2013
- [19] Collins M, Singer Y. Unsupervised models for named entity classification [J]. *Proceedings of the joint SIGDAT conference on empirical methods in natural language processing and very large corpora*. Citeseer, 1999: 100-110.
- [20] K. Nigam, R. Ghani. Analyzing the effectiveness and applicability of co-training. In: *Proceedings of the 9th ACM International Conference on Information and Knowledge Management (CIKM'00)*, McLean, VA, 2000, 86-93.
- [21] D. Angluin, P. Laird. Learning from noisy examples. *Machine Learning*, 1988, 2(4): 343-370.
- [22] S. Goldman, Y. Zhou. Enhancing supervised learning with unlabeled data. In: *Proceedings of the 17th International Conference on Machine Learning (ICML'00)*, San Francisco, CA, 2000, 327-334.
- [23] Muslea I, Minton S, Knoblock C A. Active + semi-supervised learning = robust multi-view learning [J]. *Proceedings of the 19th International Conference on Machine Learning*. Morgan Kaufmann Publishers Inc., 2002: 435-442.
- [24] Brefeld U, Scheffer T. Co-em support vector learning [J]. *Proceedings of the 21st international conference on Machine learning*. ACM, 2004: 16.
- [25] Sindhwani V, Niyogi P, Belkin M A. Co-regularization approach to semi-supervised learning with multiple views[C]// *Proceedings of ICML Workshop on Learning with Multiple Views*, 2005:74-79.
- [26] Balcan M-F, Blum A, Yang K. Co-training and expansion: Towards bridging theory and practice [J]. *Advances in Neural Information Processing Systems*, 2004: 89-96.
- [27] Y. Zhou, S. Goldman. Democratic co-learning. In: *Proceedings of the 16th IEEE International Conference on Tools with Artificial Intelligence (ICTAI'04)*, Boca Raton, FL, 2004. 594-602.
- [28] Z.-H. Zhou and M. Li. Tri-training: Exploiting unlabeled data using three classifiers. *IEEE Transactions on Knowledge and Data Engineering*, 2005, 17(11): 1529-1541.
- [29] Z.-H. Zhou, M. Li. Semi-supervised learning with co-training style algorithm. *IEEE Transactions on Knowledge and Data Engineering*, 2007, 19(11).
- [30] Wang W, Zhou Z-H. Analyzing co-training style algorithms [J]. *European Conference on Machine learning*. Springer, 2007: 454-465
- [31] S. Sun, F. Jin. Robust co-training. *Int. J. Pattern Recognit. Artif. Intell.* 25 (2011) 1113-1126.
- [32] Hotelling H. Relations between two sets of variates[J]. *Biometrika*, 1936, 28(3/4): 321-377.
- [33] Yu S, Krishnapuram B, Rómer E Rosales, et al. Bayesian Co-Training[J]. *Journal of Machine Learning Research*, 2011, 12(3):2649-2680.
- [34] Sheng S.C., Yue X.D.. A collaborative training algorithm based on rough set theory [J]. *Computer Application Research*, 2013(12):3546-3550.
- [35] Li, H. X., Zhou, H. C., Li, T. R., et al. *Decision rough set theory and its research progress* [M] Beijing: Science Press, 2011.
- [36] Guo T, Li Guiyang, Lan Xia. A graph-based semi-supervised collaborative training algorithm [J]. *Computer Engineering*, 2012, 033(009):3584-3587,3621.
- [37] Guo T, Li Guiyang, Lan Xia. A collaborative training algorithm for noise filtering based on genetic algorithm [J]. *Computer Engineering and Design*, 2014, 35(005):1807-1810.
- [38] Chen S X, Yin X X X, Yang Y J. A semi-supervised collaborative training algorithm based on code word matching and gravitational screening [J]. *Journal of Wuhan University (Information Science Edition)*, 2015(10):111-116+133.
- [39] Chen S X, Yin X X X, Yang Y J. A semi-supervised collaborative training algorithm based on code word matching and gravitational screening [J]. *Journal of Wuhan University (Information Science Edition)*, 2015(10):111-116+133.
- [40] Xiao-Yuan Jing, Fei Wu, Xiwei Dong, Shiguang Shan, and Songcan Chen. 2017. Semi-supervised multi-view correlation feature learning with application to webpage classification. In *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence (AAAI'17)*. AAAI Press, 1374-1381.
- [41] C. Chen, Y. Li, H. Qian et al., Multi-view semi-supervised learning for classification on dynamic networks, *Knowledge-Based Systems* (2020)
- [42] Farquhar J, Hardoon D, Meng H, et al. Two view learning: Svm-2k, theory and practice[J]. in *Advances in neural information processing systems*, 2005:355-362.
- [43] Akaho S. A kernel method for canonical correlation analysis[J]. *arXiv preprint cs/0609071*, 2006.
- [44] Shiliang Sun, John Shawe-Taylor. Sparse Semi-supervised Learning Using Conjugate Functions [J]. *Journal of Machine Learning Research*, 2010, 11(5):2423-2455.
- [45] Sun S. Multi-view Laplacian Support Vector Machines[C]// *Advanced Data Mining and Applications - 7th International Conference, ADMA 2011, Beijing, China, December 17-19, 2011, Proceedings, Part II*. arXiv, 2013.
- [46] Ong C S, Smola A J, Williamson R C. Learning the kernel with hyperkernels [J]. *Journal of Machine Learning Research*, 2005, 6(9): 1043-1071.
- [47] Li G, Chang K, Hoi S C H. Multiview Semi-Supervised Learning with Consensus [J]. *IEEE Transactions on Knowledge and Data Engineering*, 2012, 24(99):1-1.
- [48] Xie X, Sun S. Multi-view Laplacian twin support vector machines[J]. *Intelligent Data Analysis*, 2014, 19(4):701-712.
- [49] Zhou Z-H, Zhan D-C, Yang Q. Semi-supervised learning with very few labeled training examples[J]. in *Proceedings of the national conference on artificial intelligence*, 22(1). Menlo Park, CA; Cambridge, MA; London; AAAI Press: MIT Press; 1999, 2007:675.
- [50] Brefeld U, Christoph Büscher, Scheffer T. Multi-view Discriminative Sequential Learning[C]// *European Conference on Machine Learning*. Springer-Verlag, 2005.
- [51] Rosenberg D S, Sindhwani V, Bartlett P L, et al. Multiview point cloud kernels for semisupervised learning [Lecture Notes][J]. *Signal Processing Magazine IEEE*, 2009, 26(5):145-150.
- [52] Y. Luo, D. Tao, C. Xu, et al. Vector-valued multi-view semi-supervised learning for multi-label image classification[C]// *Twenty-seventh Aaai Conference on Artificial Intelligence*. AAAI Press, 2013.
- [53] Hu YAN, Hui HAO-TIAN, Qian LONG-HUA, et al. Bilingual collaborative relationship extraction based on machine translation[J]. *Computer Application Research*, 2015, 032(003):662-665.
- [54] Li Yegang, Huang Heyan, Shi Shumin, et al. Research on maximal noun phrase recognition based on bilingual collaborative training[J]. *Journal of Software*, 2015, 26(007):1615-1625.
- [55] Huang JY, Xiong DY. Improving the translation performance of neural machine translation systems using collaborative training[J]. *Journal of Xiamen University: Natural Science Edition*, 2019, 058(002):176-183.
- [56] Xu Y, Liu JP, Xiao YH, et al. Collaborative training-based phrase mining in e-commerce domain[J]. *Computer Engineering*, 2020(4):70-76.
- [57] Li Zhixin, Lan Danmei, Zhang Canlong, et al. A Co-Training-based method for microblog spam comment identification [J]. *Computer Engineering*, 2018, 44(07):218-224.
- [58] Du Mao-Kang, Ye Qi. Research on false comment identification based on PCA and collaborative training algorithm[J]. *Computer Simulation*, 2019, 036(002):452-457
- [59] Li Yao. Research on microblog sentiment classification based on collaborative training [D]. 2016
- [60] Yu-Ying, Chen-K, Shou-Li-Ban, et al. Sentiment analysis of user reviews based on keyword and key sentence extraction[J]. *Computer Science*, 2019, 046(010):19-26.
- [61] Li Yang. A study on collaborative training algorithm and its application in classification [D]. 2018
- [62] Yuan, Chun-Qi, Xu, Jia, Cheng, Yuan-E, et al. Semi-supervised classification of polarized SAR images based on collaborative training and integrated learning[J]. *Remote Sensing Technology and Applications*, 2017, 32(002):380-385.
- [63] Jizong Peng, Guillermo Estrada, Marco Pedersoli, Christian Desrosiers, Deep Co-Training for Semi-Supervised Image Segmentation, *Pattern Recognition* (2020)
- [64] Reinbacher C, Ruether M, Bischof H. Fast variational multi-view segmentation through backprojection of spatial constraints[J]. *Image and Vision Computing*,

- 2012, 30(11):797-807.
- [65] Yingda Xia, Dong Yang, Zhiding Yu, Fengze Liu, Jinzheng Cai, Lequan Yu, Zhuotun Zhu, Daguang Xu, Alan Yuille, Holger Roth, Uncertainty-aware multi-view co-training for semi-supervised medical image segmentation and domain adaptation, *Medical Image Analysis*, Volume 65, 2020, 101766, ISSN 1361-8415
 - [66] Y. Sun, J. Tang, W. Lei and D. He, "3D Segmentation of Pulmonary Nodules Based on Multi-View and Semi-Supervised," in *IEEE Access*, vol. 8, pp. 26457-26467, 2020
 - [67] Wang Shiku. Research on pedestrian detection based on collaborative training [D]. 2015.
 - [68] Tang Chao, Wang Wenjian, Li Wei, et al. Human behavior recognition method based on collaborative training model with multiple learners[J]. *Journal of Software*, 2015, 26(011):2939-2950.
 - [69] Wang, Chuan-An, Ge, Hua, WANG, *et al.* Analysis and identification of behavioral characteristics of "pseudo-learners" in MOOC learning [J]. *Journal of Wuhan Engineering University*, 2018.
 - [70] Liu, Xiangyang, Li, Yang, Jiang, Shuming, *et al.* Behavior recognition method based on streamwise regular co-training model[J]. *Shandong Science*, 2018, 31(001):116-120.
 - [71] Chen Tie-Ming, Zhang Kai-I, Lu Ming-Qi. A multi-view collaborative learning-based app classification method[J]. *Journal of Zhejiang University of Technology*, 2018, 46(06):591-597.
 - [72] Abdelgayed T S , Morsi W G , Sidhu T S . Fault Detection and Classification Based on Co-training of Semisupervised Machine Learning [J]. *Industrial Electronics IEEE Transactions on*, 2018, 65(2):1595-1605.
 - [73] Hassani, Mohsen & Green, James. (2019). Multi-view Co-training for MicroRNA Prediction.